Feature	Sandstorm	Furuno DRS4D-NXT	Navico HALO / BR24	Garmin xHD / GMR	Raymarine Quantum
Transport	Sandstorm	UDP multicast (Ethernet)	UDP unicast (Ethernet)	Proprietary (Ethernet)	Wi-Fi (802.11)
Spokes / rotation	Sandstorm	8,192 ticks	HALO: 2,048; BR24: 4,096	360–720	250–500
Angular resolution	Sandstorm	0.044°	HALO: 0.176°; BR24: 0.088°	0.5–1.0°	0.72–1.44°
Range bins / spoke	Sandstorm	Up to 883	512, 1,024	250–750	250–500
Wire bit depth	Sandstorm	8-bit (6-bit display)	BR24: 4-bit; HALO: 8-bit	8-bit	Lossy compressed
Encoding schemes	Sandstorm	4 adaptive (raw, RLE, delta RLE, 2-bit dual)	Raw or basic RLE	Proprietary compression	JPEG-like lossy
Compression refs	Sandstorm	Repeat cell + previous spoke	Repeat cell only	Unknown	Block-based lossy
Doppler support	Sandstorm	Native (bit 7 = D flag)	HALO: VelocityTrack; BR24: none	No	No
Dual pulse	Sandstorm	Yes (long/short interleaved)	Single pulse	Single pulse	Single pulse (FMCW)
Technology	Sandstorm	Solid-state (SSD)	Magnetron (BR24) / SSD (HALO)	Magnetron	CHIRP / FMCW

Marine Radar Echo Protocol Comparison

Based on community reverse-engineering efforts (OpenCPN radar plugins, SignalK). Competitor specifications are approximate.